

Configuring Network Routing in Linux



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Overview



Routing

- Ip
 - route show
 - route add
- Sysctl
- Netplan persistent routes



Routing: Delivery of packets to
the correct destination
network



```
$ ip route show
$ sudo ip route add 192.168.2.0/24 via 192.168.56.102
$ ip r
$ ping 192.168.2.1
$ sudo netplan apply
$ ip r
```

Network Routes

On Ubuntu 2 we can add an additional IP address, 192.168.2.1. On Ubuntu 1 we are not connected to that network and do not know how to connect to the network. Adding a route to Ubuntu 1, we can connect. This is in memory only and will be lost on startup or running netplan again

Demo



Adding Routes:

- Ubuntu 2
 - Add new IP on different network
- Ubuntu 1
 - Add transient route to remote network
 - Test
 - Reset networking



To persist network routes, we
add to netplan



```
network:
  version: 2
  renderer: networkd
  ethernets:
    enp0s8:
      addresses:
        - 192.168.56.101/24
      routes:
        - to: 192.168.2.0/24
          via: 192.168.56.102
```

Persistent Static Routes

Adding the route to netplan on Ubuntu 1 the knowledge on the remote network is retained

Demo



Persisting Static Routes:

- Ubuntu 1
- `/etc/netplan/50-vagant.yaml`
- `netplan apply`



Packet forwarding, or routing, is when a device (router) receives packets from other systems not destined for itself. Those packets are routed to the correct destination network



```
$ sysctl -ar ip_forward
$ sudo vim /etc/sysctl.conf
$ sudo sysctl -p
$ sysctl -ar ip_forward
```

Configuring Ubuntu as a Router

To forward packets for other hosts to remote networks, those networks that you are not connected to, routing must be configured. We can store the setting in the `/etc/sysctl.conf` or the `/etc/sysctl.d/`

Demo



Configuring Ubuntu as a Router

- Ubuntu 1
- sysctl.conf
- net.ipv4.ip_forward=1
- sysctl -p



Summary



Network Routing

- ip
- netplan
- sysctl



Configuring Firewalls Using IPTables

